



Study Notes

Public Goods

Introduction: Public goods

- A pure public goods is defined, as goods that one person's consumption of the goods does not reduce the amount available to others i.e. the consumption of a public good is non-rival.
- Public goods once produced, can be consumed by an additional consumer at no additional cost.
- The benefits are available to all without mutual interference
- Supplies of these goods are available to each person in community
- Thus, existence of public goods creates problems for price system
- They are underproduced or may not be produced at all in the private sector
- Therefore, either they are provided by government or government forces the people to contribute to the production of public goods, and then allow all citizens to consume them.

	Excludable	Non-Excludable
Rival	Private Goods "Typical Goods" (Clothes, Food, Flowers, etc.)	Common Goods "Common Pool Resources" (Mines, Fisheries, Forests, etc.)
Non-Rival	Club Goods "Artificially Scarce Goods" (Cable TV, Private Parks, Cinemas, etc.)	Public Goods "Collective Goods" (Air, News, Sunshine, etc.)

Figure 1. Distinction of goods

Source: Web search

Characteristics of Public goods

- There are two basic characteristics of public goods
 - ✓ **Non-rival consumption:** A good is non-rival in consumption when, with a given level of production, consumption by one person need not diminish the quantity consumed by anyone else.
 - A number of people may simultaneously consume the same good.
 - The cost of extending the service or providing the good to another person is zero
 - ✓ **Non-exclusion:** Non-exclusion means that it is impossible to confine the benefits of the good (once produced) to selected persons
 - It is impossible to exclude anyone from enjoying the benefits of a public good
 - A person, whether he pays for it or not, will benefit from the production of the public good (free-rider)
- Some other features of public goods includes Non-Rejectable, free-rider problem
 - **Indivisibility:** All consumers can get equal amount benefit from public goods provided they

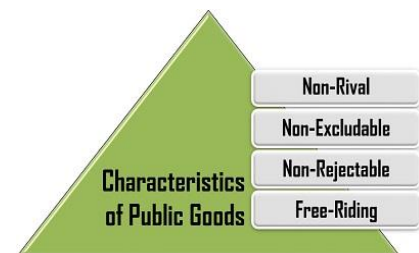


Figure 2. Characteristics of Public goods

Source: Web search



Figure 3. Examples of Public goods

Source: Web search

are capable of consuming it.

- **Externality:** The benefits accrued from public good are external to the consumer. They are subject to the principle of consumer sovereignty. They are produced on the basis of individual preference, but the satisfaction derived from public goods by an individual consumer is independent of his own contribution. This is so because, the cost met by government through budget
- **Benefit Obscure:** It is very difficult to realise the benefit from public good
- **Payee and Beneficiary not identical:** The same who pays for it to make its provision certainly are not a real beneficiary
- Examples
 - ✓ National Defence
 - ✓ Roads
 - ✓ Government parks
 - ✓ Street lights
 - ✓ Traffic lights

Market Failure of Public Good

Market failure is a situation where there is inefficient distribution of goods and services in the free market.

Optimal Quantity

- The demand curve is the marginal social benefit curve
- The supply curve is the marginal social cost curve
- The optimal quantity of the Public good to be produced is given by $MSB = MSC$ (Fig 4.)
- i.e. Marginal Social benefit = Marginal Social Cost

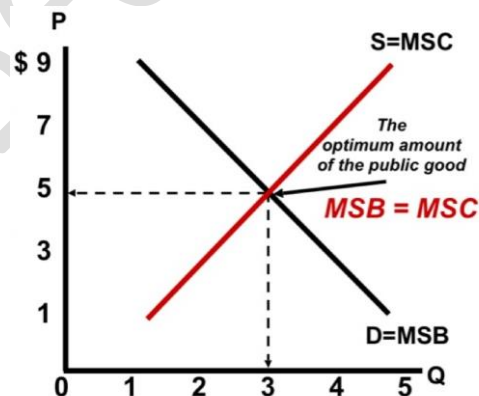


Figure 4. Optimal amount of Public goods
Source: Web search

Underproduction of Public Goods

- Presence of **free riders** makes it difficult or impossible for market to provide public goods efficiently.
- Therefore, the quantity Q , produced otherwise is reduced to Q_1 . (Fig 5.)
- Q_1Q is the underproduction
- And ABC represents the welfare loss
- In this case private arrangements are usually ineffective
- For efficient production, public good must be subsidized or provided by governments
- However, in the real world application of public goods theory has certain problems
- Many public goods are successfully produced in the private sector

- Many of the goods actually produced do not correspond to economic definition of public goods.
- Thus, public goods certainly exist but production in the public sector neither necessary nor sufficient for the efficient production of public goods

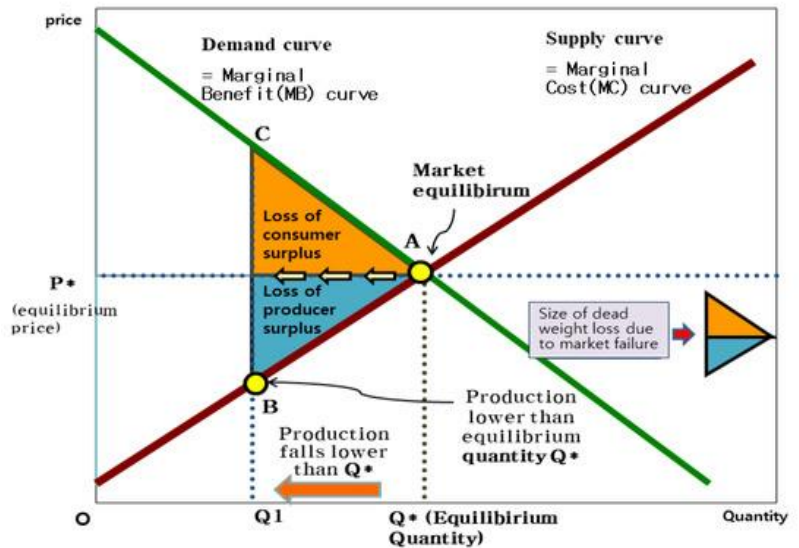


Figure 5. Market failure of Public goods

Source: Web search